

taining high temperature difference between two ends of the device.

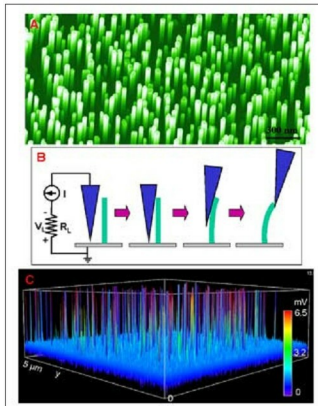
Vibration-based energy generator. Scientists have built vibration-based generators using three types of electromechanical transducers—electromagnetic, electrostatic and piezoelectric. The electromagnetic microgenerator utilises a moving magnet or coil for inducing and alternating electric current in closed circuit.

Piezoelectric vibration-based generator is based on piezoelectricity, which is the ability of certain crystals to generate a voltage in response to applied mechanical stress. The studies are based on aligned ZnO nanowires (NWs) grown on a conductive solid substrate. The measurement work is performed by an atomic force microscope (AFM) using a silicon tip coated with platinum film. The deflection of NWs by AFM tip creates a strain field with the outer surface being tensile and the inner surface compressive. As a result, across the width of NW at the top end, a voltage drop occurs, with positive voltage on tensile side and negative voltage on compressive side.

Advantages of ZnO NWs are:

1. NWs can be subjected to large deformation, hence they can be used for flexible electronics.
2. ZnO is bio-compatible material, so it can be implanted in human body.
3. ZnO is an environmentally green material.

Direct current nanogenerator driven



A scanning electron microscope image (top) shows an array of zinc oxide nanowires. Middle image shows a schematic of how an AFM tip was used to bend nanowires to produce current. Bottom image depicts output voltages produced by the array as it is scanned by the probe (Image courtesy: Zhong Lin Wang)

en by ultrasonic waves. Here waves or vibrations from the environment are used to provide electricity in nanogenerator. Hence, nanodevices can be operated independently and wirelessly.

Flexible power generator built on polymer substrate. The semiconductor substrates used for growing ZnO NWs are hard, brittle and cannot be used in the areas that require foldable or flexible power source, such as implantable bio sensors in muscles or joints, and power generator built in walking shoes. It is necessary to use conductive and bio-compatible polymers to harness energy from body movement, gestures, respiration and locomotion.

Fibre-based nanogenerators. In order to build on flexible and foldable substrate, the design of nanogenerator is based on the mechanism of utilising a zigzag electrode by an array of metal wires. By brushing the metal nanowire arrays across the ZnO nanowire arrays, the metal wires act like an array of AFM tips that induce the mechanical deformation to the ZnO nanowires and then collect the charges.

Rotating magnetic particle-based nanogenerator. In such devices, particles are put into motion by imposed ferrofluid or air driven flows generating electric power in surrounding nanocoil windings by changing the magnetic flux with time through nanocoils.

The above described approaches are very useful in biomedical science. Among all the methods described above, piezoelectric is one of

the most exciting fields in nanotechnology.

Self-powered nanodevices offer monitoring of blood pressure and blood sugar levels. These can be used to diagnose cancer cells or wirelessly measure fluid pressure in brain. A nanogenerator could also be used to remotely sense gas and chemicals and track down animal migration in the field of environmental science. We can obtain energy from mechanical movements of the body, like swinging arms and stretching legs, for charging batteries of personal electronics.

A nanogenerator may harvest energy from sound waves, ultrasound waves, wind and airflow, mechanical vibrations or thermal noise. It may also help in harvesting and recycling wasted energy, such as energy created by pressure change in vehicle tyres and a moving car's mechanical vibrations. ●

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